RETAIL SALES DATA ANALYTICS PROJECT REPORT

Retail Sales Data Analysis Using SQL

Submitted by

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ABSTRACT

Retail organizations generate large volumes of transactional data through daily sales activities. Analyzing this data helps businesses understand customer purchasing behavior, identify sales trends, and improve decision-making processes.

This project focuses on analyzing retail sales transaction data using Structured Query Language (SQL). The dataset contains customer transaction details such as customer ID, transaction date, and transaction amount. Various SQL queries are used to calculate total sales, identify top customers, and analyze customer spending patterns.

The results of this analysis provide useful insights into customer behavior and business performance. Retail companies can use these insights to improve marketing strategies, enhance customer relationships, and increase revenue.

1. INTRODUCTION

In the modern business environment, data analytics has become an essential tool for organizations to gain insights from large datasets. Retail businesses generate a large amount of data through daily transactions, including information about customers, purchases, and sales revenue.

Analyzing retail sales data helps organizations understand customer purchasing patterns and identify key factors affecting sales performance. SQL is widely used for analyzing structured data stored in relational databases.

This project uses SQL queries to analyze retail sales transaction data and generate meaningful insights that can support business decision-making.

2. LITERATURE REVIEW

Several studies have highlighted the importance of data analytics in retail businesses. Retail companies collect large volumes of data that can be analyzed to understand customer behavior and improve sales strategies.

Researchers have shown that analyzing transactional data helps identify high-value customers, predict sales trends, and improve marketing strategies. SQL-based data analysis is widely used because it allows efficient querying and aggregation of large datasets stored in relational databases.

Retail analytics techniques are commonly used for customer segmentation, demand forecasting, and business performance analysis.

3. PROBLEM STATEMENT

Retail organizations store a large amount of sales data but may not always analyze it effectively. Without proper analysis, businesses cannot fully understand customer purchasing behavior or identify opportunities to improve sales performance.

The objective of this project is to analyze retail transaction data using SQL queries and generate meaningful insights about customer spending patterns and overall sales performance.

4. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To analyze retail sales transaction data
- To calculate total sales revenue
- To identify top customers based on spending

• To analyze transaction frequency
• To generate useful business insights from the dataset

5. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in this project:

- ❖ MySQL – Database management system used to store and manage the dataset.
- ❖ SQL – Structured Query Language used for performing data analysis.
- ❖ MySQL Workbench – Tool used to execute SQL queries and manage the database.
- ❖ Microsoft Excel – Used to create charts and visualize analysis results.

6. DATASET DESCRIPTION

The dataset used in this project contains retail transaction data. Each row in the dataset represents a purchase made by a customer.

The dataset includes the following attributes:

- ❖ Customer_ID – Unique identifier assigned to each customer.
- ❖ Transaction_Date – Date on which the transaction occurred.
- ❖ Transaction_Amount – Amount spent by the customer in the transaction.

This dataset is imported into a MySQL database table for further analysis.

7. METHODOLOGY

The project methodology consists of several steps.

Step 1 – Dataset Collection
The retail sales dataset was collected for analysis.

Step 2 – Data Preparation
The dataset was cleaned and verified to ensure correct formatting.

Step 3 – Database Creation
A database table was created in MySQL to store the dataset.

Step 4 – Data Import
The dataset was imported into the MySQL table.

Step 5 – SQL Query Execution
SQL queries were executed to analyze the dataset.

Step 6 – Result Analysis
The results of the SQL queries were analyzed to generate insights.

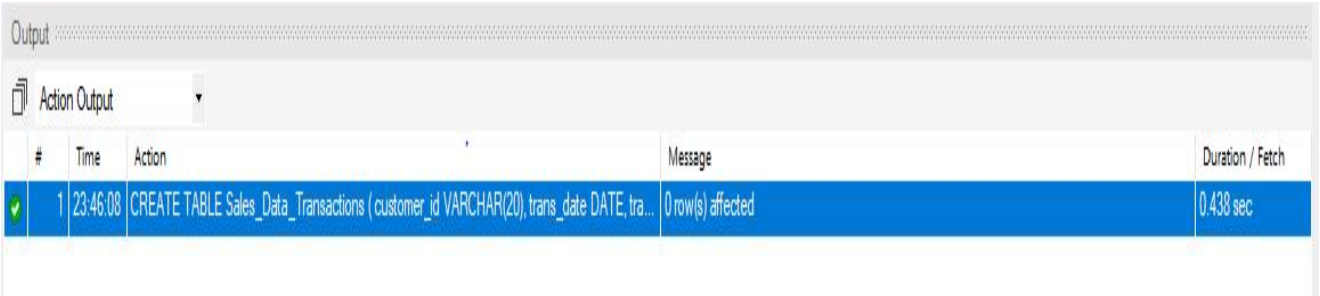
Step 7 – Visualization
Charts were created to represent the results visually.

8. DATABASE DESIGN

The following SQL command was used to create the database table.

```
CREATE TABLE Sales_Data_Transactions (
customer_id VARCHAR(20),
trans_date DATE,
tran_amount INT
);
```

Result Screenshot:



Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	23:46:08	CREATE TABLE Sales_Data_Transactions (customer_id VARCHAR(20), trans_date DATE, tra...	0 row(s) affected	0.438 sec

Explanation:

This table stores retail transaction information including customer ID, transaction date, and transaction amount.

9. DATA LOADING

The dataset was imported into MySQL using the data import wizard available in MySQL Workbench. After importing the dataset, the data was stored in the Sales_Data_Transactions table and made ready for analysis.

```
CREATE DATABASE retail_sales_db ;

USE retail_Sales_db ;

CREATE TABLE Sales_Data_Transactions (

   customer_id VARCHAR(255),

   trans_date VARCHAR(255),

   tran_amount INT
```

```
);

CREATE TABLE Sales_Data_Response (

   customer_id VARCHAR(255),

   response INT

);

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Retail_Data_Transactions.txt'

INTO TABLE Sales_Data_Transactions

FIELDS TERMINATED BY ','

LINES TERMINATED BY '\n'

IGNORE 1 ROWS;

SELECT * FROM  Sales_Data_Transactions LIMIT 10;
```

Result Screenshot:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customer_id	trans_date	tran_amount
CS5295	11-Feb-13	35
CS4768	15-Mar-15	39
CS2122	26-Feb-13	52
CS1217	16-Nov-11	99
CS1850	20-Nov-13	78
CS5539	26-Mar-14	81
CS2724	06-Feb-12	93
CS5902	30-Jan-15	89

10. SQL QUERIES AND ANALYSIS

Query 1 – Total Sales

```
SELECT SUM(tran_amount) AS total_sales
FROM Sales_Data_Transactions;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_sales
8123989

Explanation

This query calculates the total sales revenue generated from all transactions in the dataset.

Query 2 – Total Transactions

```
SELECT COUNT(*) AS total_transactions
FROM Sales_Data_Transactions;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_transactions
125000

Explanation

This query counts the total number of transactions recorded in the dataset.

Query 3 – Customer Spending

```
SELECT customer_id, SUM(tran_amount) AS total_spent
FROM Sales_Data_Transactions
GROUP BY customer_id;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customer_id	total_spent
CS5295	1637
CS4768	1721
CS2122	1648
CS1217	1431
CS1850	1908
CS5539	1354
CS2724	1166
CS5902	922

Explanation

This query calculates the total amount spent by each customer.

Query 4 – Top Customers

```
SELECT customer_id, SUM(tran_amount) AS total_spent
FROM Sales_Data_Transactions
GROUP BY customer_id
ORDER BY total_spent DESC
LIMIT 5;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customer_id	total_spent
CS4424	2933
CS4320	2647
CS5752	2612
CS4660	2527
CS3799	2513

Explanation

This query identifies the top five customers based on total spending.

Query 5 – Average Transaction

```
SELECT AVG(tran_amount)
FROM Sales_Data_Transactions;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	AVG(tran_amount)
▶	64.9919

Explanation

This query calculates the average transaction amount.

Query 6 – Highest Transaction

```
SELECT MAX(tran_amount)
FROM Sales_Data_Transactions;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	MAX(tran_amount)
▶	105

Explanation

This query identifies the highest transaction amount recorded in the dataset.

Query 7 – Lowest Transaction

```
SELECT MIN(tran_amount)
FROM Sales_Data_Transactions;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	MIN(tran_amount)
▶	10

Explanation

This query identifies the lowest transaction amount.

Query 8 – Transactions per Customer

```
SELECT customer_id, COUNT(*)
FROM Sales_Data_Transactions
GROUP BY customer_id;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	customer_id	COUNT(*)
▶	CS5295	25
	CS4768	26
	CS2122	23
	CS1217	20
	CS1850	24
	CS5539	18
	CS2724	14
	CS5902	13

Explanation

This query calculates the number of transactions performed by each customer.

Query 9 – Monthly Sales

```
SELECT MONTH(trans_date), SUM(tran_amount)
FROM Sales_Data_Transactions
GROUP BY MONTH(trans_date);
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MONTH(trans_date)	SUM(tran_amount)
NULL	8123989

Explanation

This query calculates the total sales for each month.

Query 10 – High Value Customers

```
SELECT customer_id, SUM(tran_amount)
FROM Sales_Data_Transactions
GROUP BY customer_id
HAVING SUM(tran_amount) > 1000;
```

Result Screenshot

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customer_id	SUM(tran_amount)
CS5295	1637
CS4768	1721
CS2122	1648
CS1217	1431
CS1850	1908
CS5539	1354
CS2724	1166
CS6040	1001

Explanation
This query identifies customers who spent more than 1000 in total.

11. RESULTS AND DISCUSSION

The SQL queries generated several insights regarding retail sales performance. The analysis showed the overall revenue generated from customer transactions and helped identify customers who contribute significantly to total sales.

Understanding customer spending patterns helps businesses design better marketing strategies and improve customer engagement.

12. DATA VISUALIZATION

Charts were created to visualize the results of the analysis.

Examples include:

Bar Chart – Customer spending comparison
Line Chart – Monthly sales trend
Pie Chart – Customer contribution to total sales

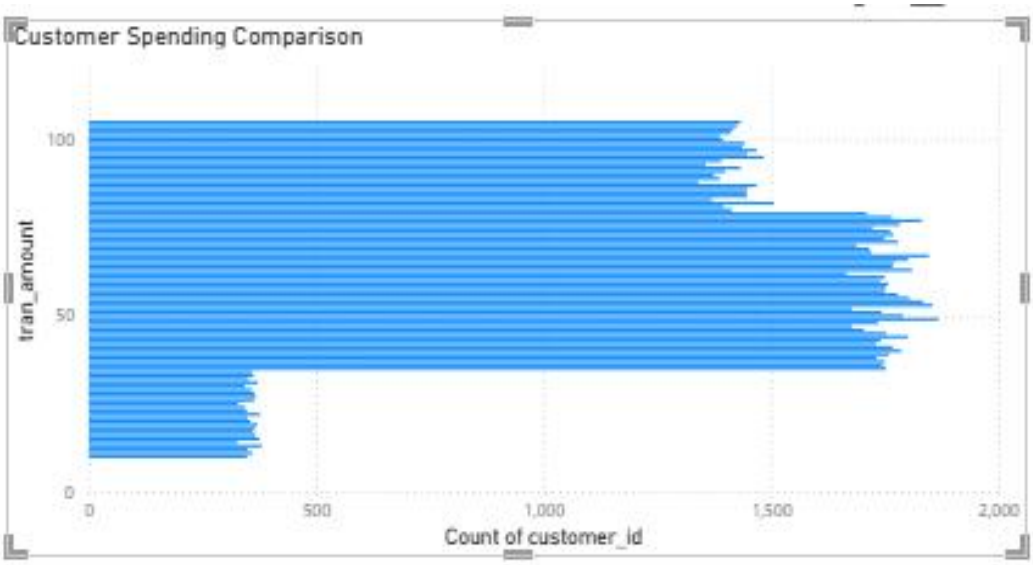


Fig : Customer Spending Bar Chart

The bar chart represents the total spending made by each customer in the dataset. Each bar corresponds to a specific customer and shows the total transaction amount contributed by that customer. This visualization helps identify customers who contribute the highest revenue to the business. By analyzing customer spending patterns, businesses can recognize high-value customers and develop strategies to retain them through loyalty programs or special offers.

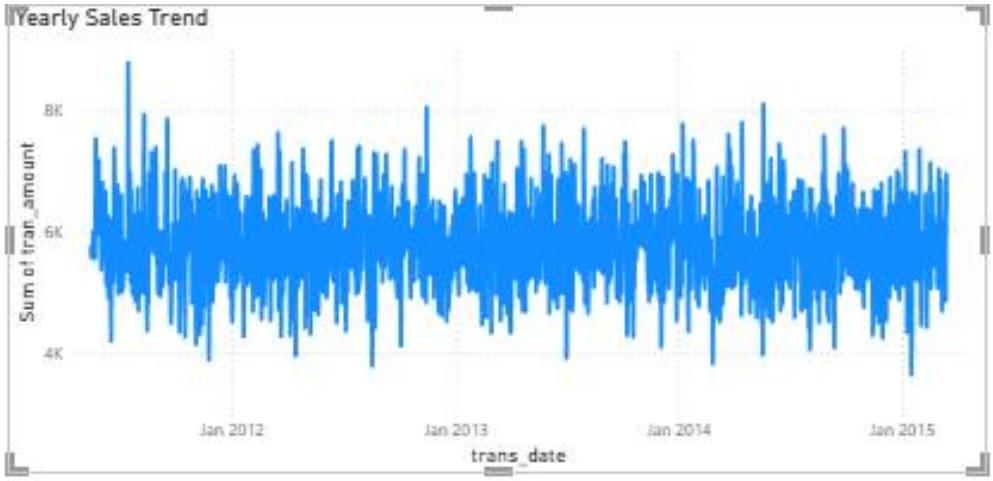


Fig: Yearly Sales Trend

The line chart illustrates the yearly sales trend based on the transaction data. It shows how the total sales revenue changes across different years. The horizontal axis represents the years, while the vertical axis represents the total transaction amount. This visualization helps analyze the growth or decline in sales over time. Businesses can use this information to understand long-term sales performance and make better strategic decisions for future growth.

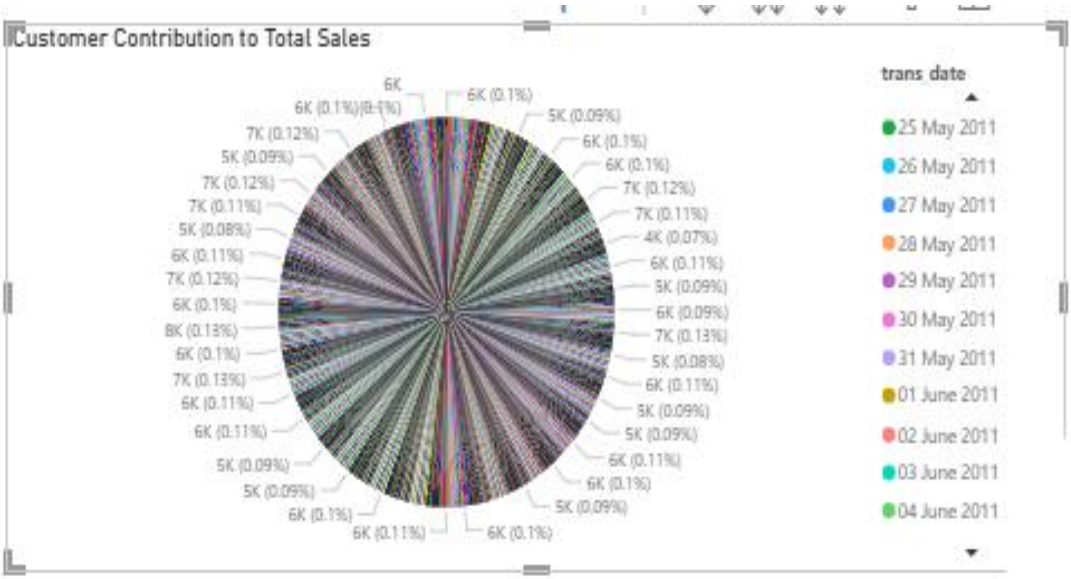


Fig: Customer Contribution to Total Sales

The pie chart shows the percentage contribution of each customer to the total sales revenue. Each slice of the pie represents a customer's share of the total transaction amount. This visualization helps identify the customers who contribute significantly to the overall revenue. Businesses can use this information to focus on important customers and improve customer relationship strategies.

13. INSIGHTS

The analysis revealed several useful insights.

• A small group of customers contributes significantly to total revenue.
• Customer spending patterns vary across transactions.
• Yearly sales analysis shows variations in sales performance.

These insights can help businesses develop better strategies to increase sales and improve customer retention.

14. BUSINESS RECOMMENDATIONS

Retail businesses should focus on high-value customers and offer loyalty programs to retain them. Data analytics should be used regularly to monitor sales trends and customer behavior.

Businesses can also use these insights to design targeted marketing campaigns.

15. CONCLUSION

This project demonstrated how SQL queries can be used to analyze retail sales transaction data. The analysis helped identify important sales metrics and customer spending patterns.

Data analytics plays an important role in helping businesses make informed decisions and improve overall performance.

16. FUTURE ENHANCEMENTS

The project can be further improved by:

• Using larger datasets
• Implementing Power BI dashboards
• Applying machine learning techniques for sales prediction
• Performing customer segmentation analysis

17. REFERENCES

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